

Fundamentals of AI and KR - Module 3

1. Introduction to uncertainty and probabilistic reasoning

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Notice

Credits

The present slides are largely an adaptation of existing material, including:

- slides from [Russel & Norvig](#)
- slides by [Daphne Koller](#) on Probabilistic Graphical Models
- slides by Fabrizio Riguzzi on [Data Mining and Analytics](#)

I am especially grateful to these authors.

Downloading and sharing

A copy of these slides can be downloaded from [virtuale](#) and stored for personal use only. Please do not redistribute.

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- Acting under uncertainty
- Basic probability notation
- Inference using full joint distributions

Introduction and logistics

Introduction

Problem-solving logical agents

- Consider 3 switches in a switchboard: A, B, and C. When A is on, B is also on. When C is on, B is off. If both A and C are on, the switchboard melts down. *Can that ever happen?*

Introduction

Problem-solving logical agents

- Consider 3 switches in a switchboard: A, B, and C. When A is on, B is also on. When C is on, B is off. If both A and C are on, the switchboard melts down. *Can that ever happen?*
- There are 12 coins distributed in piles of 4, 1, and 7 coins. We can move coins from a pile A to a pile B, but only by doubling the coins in B. *Can we distribute coins evenly among the three piles?*

Introduction

Problem-solving logical agents

- Consider 3 switches in a switchboard: A, B, and C. When A is on, B is also on. When C is on, B is off. If both A and C are on, the switchboard melts down. *Can that ever happen?*
- There are 12 coins distributed in piles of 4, 1, and 7 coins. We can move coins from a pile A to a pile B, but only by doubling the coins in B. *Can we distribute coins evenly among the three piles?*
- We're **guests** on a TV game show. We stand in front of three closed doors. A **prize** hides behind one of them. We choose the door on the left. At this point, the **host**, who knows where prize is, opens the middle door, to reveal it is empty. We are offered to modify our choice. *Should we?*



Handling uncertainty

Agents may need to handle uncertainty due to...

- partial observability
- nondeterminism
- a combination of both

Problem-solving and logical agents keep a belief state and generate a contingency plan. However...

- large and complex belief-state representations
- arbitrarily large contingency plan
- there may be no plan guaranteed to achieve the goal

Application areas

- Robotics
- Medical diagnosis
- Troubleshooting
- Decision-making
- Risk assessment
- Automated monitoring
- Predictions
- Image and speech synthesis/recognition
- Computational biology
- Economics
- ...

Topics

- Basic probability notation
- Inference using full joint distributions
- Independence
- Bayesian network representation
- Constructing Bayesian networks
- Exact and approximate inference
- Simple case studies

Learning resources

- Course slides

- Textbook

- **Artificial Intelligence. A Modern Approach**, by Stuart Russel and Peter Norvig. Pearson Education. 2nd, 3rd Ed. Chapters 13 & 14, or 4th Ed. Chapters 12 & 13.

- Additional reading

- **Probabilistic Graphical Models. Principles and Techniques**, by Daphne Koller and Nir Friedman. MIT Press.
- **Foundations of Probabilistic Logic Programming. Languages, Semantics, Inference and Learning**, by Fabrizio Riguzzi. River Publishers.

- Software

- PGM Python library: [pgmpy](#)



Exam

- Two alternative possibilities: written exam or project
- Written exam:
 - Questions and/or exercises of the type seen in class
 - 4 dates per academic year
- Mini-project:
 - Implementation of simple case study in pgmpy or other library
 - Upload: ipython notebook and PDF report
 - Oral exam to present work done and answer questions
 - 3 discussion periods per academic year
- Grades for M3 on an 11-point scale (11 is A+)
 - Other modules' max points: 32 (M1) and 21 (M2)
- Final grade is:

$$\frac{\sum_{n=1}^3 \text{grade}(\text{Module } n)}{2}$$

- Round up between 18 and 30, but 30L only if $\frac{\sum_i \text{grade}(M_i)}{2} \geq 31.0$

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Acting under uncertainty



Uncertainty

We need to reach the airport on time. Let action A_t = leave for airport t minutes before flight. *Will A_t get me there on time?*

Problems:

- ① partial observability (road state, other drivers' plans, etc.)
- ② noisy sensors (traffic reports)
- ③ uncertainty in action outcomes (flat tire, etc.)
- ④ immense complexity of modelling and predicting traffic

Hence a purely logical approach either

- ① risks falsehood: " A_{25} will get me there on time," or
- ② leads to conclusions that are too weak for decision making: " A_{25} will get me there on time if there's no accident on the bridge, and it doesn't rain and my tires remain intact etc etc."

(A_{1440} might reasonably be said to get me there on time but I'd have to stay overnight in the airport ...)



Methods for handling uncertainty (M2)

Default or nonmonotonic logic

- Assume my car does not have a flat tire
- Assume A_{25} works unless contradicted by evidence

What assumptions are reasonable?

Rule-based systems with fudge factors

- $A_{25} \mapsto_{0.3} AtAirportOnTime$
- rules for causal reasoning: $FaultyPowerCord \mapsto_{0.99} DisplayOff$
- rules for diagnostic reasoning: $DisplayOff \mapsto_{0.7} SleepMode$

Issues with locality, e.g., how can 0.3 account for "all" the evidence?

Issues with combination, e.g., $FaultyPowerCord$ causes $SleepMode$?



Methods for handling uncertainty

Probability

Given the available evidence, A_{25} will get me there on time with probability 0.04

Remark. Fuzzy logic handles degree of truth NOT uncertainty e.g.,

- *TrafficCongested* is true to degree 0.8



Probability

Probabilistic assertions summarize uncertainty due to:

- **laziness**: failure to enumerate exceptions, qualifications, etc.
- **ignorance**: lack of relevant facts, initial conditions, etc.

Subjective or Bayesian probability:

Probabilities relate propositions to one's own state of knowledge

- e.g., $P(A_{25} | \text{no reported accidents}) = 0.06$

These are not claims of a “probabilistic tendency” in the current situation
(but might be learned from past experience of similar situations)

Probabilities of propositions change with new evidence:

- e.g., $P(A_{25} | \text{no reported accidents, 5 a.m.}) = 0.15$



Making decisions under uncertainty

Suppose I believe the following:

$$P(A_{25} \text{ gets me there on time} | \dots) = 0.04$$

$$P(A_{90} \text{ gets me there on time} | \dots) = 0.70$$

$$P(A_{120} \text{ gets me there on time} | \dots) = 0.95$$

$$P(A_{1440} \text{ gets me there on time} | \dots) = 0.9999$$

Which action to choose?

- Depends on my **preferences** for missing flight vs. airport cuisine, etc.
- Being **Rational** means following **Maximum Expected Utility** principle
- **Utility theory** is used to represent and infer preferences
- **Decision theory** = utility theory + probability theory

Basic probability notation



Probability basics

Consider the assertions about possible worlds

- Logical assertions say which worlds are ruled out
- Probabilistic assertions say how probable they are

Sample space and events

The set of all possible worlds is called the **sample space**, denoted Ω . Any subset $A \subseteq \Omega$ is an **event**. Any element $\omega \in \Omega$ is called a **sample point/possible world/atomic event**

e.g., 6 possible rolls of a die; die roll < 4 ; die roll $= 3$
Dado



Probability basics

Probability space

A **probability space** or **probability model** is a sample space with an assignment $P(\omega)$ for every $\omega \in \Omega$ s.t.

- $0 \leq P(\omega) \leq 1$
- $\sum_{\omega} P(\omega) = 1$

Accordingly,

$$P(A) = \sum_{\omega \in A} P(\omega)$$

e.g., $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$.

$$P(\text{die roll} < 4) = P(1) + P(2) + P(3) = 1/6 + 1/6 + 1/6 = 1/2$$



Random variables

Random variables

A **random variable** is **a function** from sample points to some range, e.g., the reals or Booleans.

e.g., $Odd(1) = true$.

Probability distribution

P induces a **probability distribution** for any r.v. X :

$$P(X = x_i) = \sum_{\omega: X(\omega) = x_i} P(\omega)$$

e.g., $P(Odd = true) = P(1) + P(3) + P(5) = 1/6 + 1/6 + 1/6 = 1/2$



Propositions

Think of a proposition as the event where the proposition is true.
e.g., given Boolean random variables A and B :

- event a = set of sample points where $A(\omega) = \text{true}$
- event $\neg a$ = set of sample points where $A(\omega) = \text{false}$
- event $a \wedge b$ = points where $A(\omega) = \text{true}$ and $B(\omega) = \text{true}$

With Boolean variables, sample point = propositional logic model
e.g., $A = \text{true}$, $B = \text{false}$, or $a \wedge \neg b$.

Proposition = disjunction of atomic events in which it is true

e.g., $(a \vee b) \equiv (\neg a \wedge b) \vee (a \wedge \neg b) \vee (a \wedge b)$
 $\Rightarrow P(a \vee b) = P(\neg a \wedge b) + P(a \wedge \neg b) + P(a \wedge b)$



Syntax for propositions

- **Propositional** or **Boolean** random variables
e.g., *Cavity* (do I have a cavity?)
Cavity = true is a proposition, also written *cavity*
- **Discrete** random variables (**finite** or **infinite**)
e.g., *Weather* is one of *⟨sunny, rain, cloudy, snow⟩*
Weather = rain is a proposition
Values must be exhaustive and mutually exclusive
- **Continuous** random variables (**bounded** or **unbounded**)
e.g., *Temp = 21.6*; *Temp < 22.0*
- Arbitrary Boolean combinations of basic propositions



Prior probability

Prior probability

Prior or unconditional probabilities of propositions correspond to belief prior to arrival of any (new) evidence

e.g., $P(\text{Cavity} = \text{true}) = 0.1$ and $P(\text{Weather} = \text{sunny}) = 0.72$

Probability distribution

A probability distribution gives values for all possible assignments.

e.g. $\mathbf{P}(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, i.e., sums to 1)



Joint Probability Distribution

Joint Probability Distribution

The **Joint Probability Distribution** for a set of r.v.s gives the probability of every atomic event on those r.v.s (i.e., every sample point)

e.g. $\mathbf{P}(\textit{Weather}, \textit{Cavity})$ = a 4×2 matrix of values:

<i>Weather</i> =	<i>sunny</i>	<i>rain</i>	<i>cloudy</i>	<i>snow</i>
<i>Cavity</i> = <i>true</i>	0.144	0.02	0.016	0.02
<i>Cavity</i> = <i>false</i>	0.576	0.08	0.064	0.08

Every question about a domain can be answered by the joint distribution because every event is a sum of sample points



Probability for continuous variables

Probability density function

A function $p : \mathbb{R} \rightarrow \mathbb{R}$ is a **probability density function (pdf)** for X if it is a nonnegative integrable function s.t.

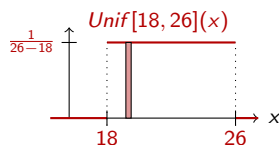
$$\int_{\text{Val}(X)} p(x) dx = 1.$$

A common **pdf** is the **Uniform distribution**

$$p(x) = \text{Unif}[a, b](x) = \begin{cases} \frac{1}{b-a} & b \geq x \geq a \\ 0 & \text{otherwise} \end{cases}$$

What $P(X = 20.5) = 0.125$ really means is:

$$\lim_{dx \rightarrow 0} \frac{P(20.5 \leq X \leq 20.5 + dx)}{dx} = 0.125$$





Probability for continuous variables

Probability density function

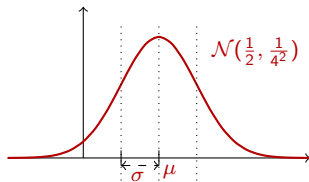
A function $p : \mathbb{R} \rightarrow \mathbb{R}$ is a **probability density function (pdf)** for X if it is a nonnegative integrable function s.t.

$$\int_{\text{Val}(X)} p(x) dx = 1.$$

Another common **pdf** is the **Gaussian** (Normal) distribution

$$\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Standard Gaussian: $\mathcal{N}(0, 1) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$





Conditional probability

With respect to prior probabilities $P(X)$, **conditional** or **posterior probabilities** $P(X|Evidence)$ represent a more informed distribution in the light of the (new) *Evidence*.

e.g., $P(cavity|toothache) = 0.8$:

i.e., **given that** *toothache* **is all I know**

NOT “if *toothache* then 80% chance of *cavity*”

(Notation for **sets** of conditional distributions:

$P(Cavity|Toothache)$ is a 2-element vector of 2-element vectors)

Bold P indicates the distribution and not just the probability



Conditional probability

$$P(\text{cavity}|\text{toothache}) = 0.8$$

If we know more, e.g., *cavity* is also given, then we have

$$P(\text{cavity}|\text{toothache}, \text{cavity}) = 1$$

Note: the less specific belief remains valid after more evidence arrives, but is not always useful

New evidence may be irrelevant, allowing simplification, e.g.,

$$P(\text{cavity}|\text{toothache}, 49ersWin) = P(\text{cavity}|\text{toothache}) = 0.8$$

This kind of inference, sanctioned by domain knowledge, is crucial



Conditional probability

Definition of conditional probability: $P(a|b) = \frac{P(a \wedge b)}{P(b)}$ if $P(b) \neq 0$

Product rule gives an alternative formulation:

$$P(a \wedge b) = P(a|b)P(b) = P(b|a)P(a)$$

A general version holds for whole distributions, e.g.,

$$\mathbf{P}(\textit{Weather}, \textit{Cavity}) = \mathbf{P}(\textit{Weather} | \textit{Cavity}) \mathbf{P}(\textit{Cavity})$$

(View as a 4×2 set of equations, not matrix mult.)

<i>Weather</i> =	sunny	rain	cloudy	snow
<i>Cavity</i> = <i>true</i>	0.144	0.02	0.016	0.02
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Conditional probability

Chain rule is derived by successive application of product rule:

$$\begin{aligned}\mathbf{P}(X_1, \dots, X_n) &= \mathbf{P}(X_1, \dots, X_{n-1}) \mathbf{P}(X_n | X_1, \dots, X_{n-1}) \\ &= \mathbf{P}(X_1, \dots, X_{n-2}) \mathbf{P}(X_{n-1} | X_1, \dots, X_{n-2}) \mathbf{P}(X_n | X_1, \dots, X_{n-1}) \\ &= \dots \\ &= \prod_{i=1}^n \mathbf{P}(X_i | X_1, \dots, X_{i-1})\end{aligned}$$

In this way we decompose a complex probability as a multiplication of simpler ones and in a single table you obtain a full-joint distribution

Inference using full joint distributions



Inference by enumeration

Start with the joint distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

$P(\text{toothache})$

For any proposition ϕ , sum the atomic events where it is true:

$$P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$$

e.g. $\phi = \text{toothache}$



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$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$



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e.g. $\phi = \text{cavity} \vee \text{toothache}$



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For any proposition ϕ , sum the atomic events where it is true:

$$P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$$

$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

$$P(\text{cavity} \vee \text{toothache}) = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28$$



Inference by enumeration

Start with the joint distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

Can also compute conditional probabilities:

$$P(\neg\text{cavity}|\text{toothache}) = \frac{P(\neg\text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$



Inference by enumeration

Start with the joint distribution:

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Can also compute conditional probabilities:

$$\begin{aligned}
 P(\neg \text{cavity} | \text{toothache}) &= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\
 &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4
 \end{aligned}$$



Normalization

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
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Denominator can be viewed as a **normalization constant** α

$$\mathbf{P}(\textit{Cavity}|\textit{toothache}) = \alpha \mathbf{P}(\textit{Cavity}, \textit{toothache})$$

alpha so that the sum leads up to 1



Normalization

	<i>toothache</i>		$\neg$ <i>toothache</i>	
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Denominator can be viewed as a ^{Testo}normalization constant α

$$\begin{aligned}
 \mathbf{P}(Cavity|toothache) &= \alpha \mathbf{P}(Cavity, toothache) \\
 &= \alpha [\mathbf{P}(Cavity, toothache, catch) + \mathbf{P}(Cavity, toothache, \neg catch)] \\
 &= \alpha [\langle 0.108, 0.016 \rangle + \langle 0.012, 0.064 \rangle] \\
 &= \alpha \langle 0.12, 0.08 \rangle = \langle 0.6, 0.4 \rangle
 \end{aligned}$$

$\alpha * (0.12 + 0.08) = 1 \implies \alpha = 5$



Normalization

	<i>toothache</i>		$\neg$ <i>toothache</i>	
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 &= \alpha [\langle 0.108, 0.016 \rangle + \langle 0.012, 0.064 \rangle] \\
 &= \alpha \langle 0.12, 0.08 \rangle = \langle 0.6, 0.4 \rangle
 \end{aligned}$$

General idea: compute distribution on **query variable** by fixing **evidence variables** and summing over **hidden variables**

Common terminology for operations on CPDs

<i>Weather =</i>	<i>sunny</i>	<i>rain</i>	<i>cloudy</i>	<i>snow</i>
<i>Cavity = true</i>	0.144	0.02	0.016	0.02
<i>Cavity = false</i>	0.576	0.08	0.064	0.08

- **Marginalization or Summing Out**
e.g., $P(\textit{Weather} = \textit{sunny})$

Common terminology for operations on CPDs

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- Marginalization or Summing Out

e.g., $P(\textit{Weather} = \textit{sunny})$

- Conditioning

e.g., condition on $\textit{Weather} = \textit{sunny}$: $P(\textit{Cavity} | \textit{Weather} = \textit{sunny})$

⇒ reduction and renormalization



Probability queries

Probability query

A probability query $P(\mathbf{Y}|\mathbf{e})$ defines the posterior joint distribution of a set of query variables $\mathbf{Y}$ given specific values $\mathbf{e}$ for some evidence variables.

We thus have three sets of r.v.s: query variables $\mathbf{Y}$, evidence variables $\mathbf{E}$, and hidden variables $\mathbf{H}$ (all else).

In principle, one could answer the query by summing out.

$$P(\mathbf{Y}|\mathbf{E}=\mathbf{e}) = \alpha P(\mathbf{Y}, \mathbf{E}=\mathbf{e}) = \dots$$



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In principle, one could answer the query by summing out.

$$P(\mathbf{Y}|\mathbf{E}=\mathbf{e}) = \alpha P(\mathbf{Y}, \mathbf{E}=\mathbf{e}) = \alpha \sum_{\mathbf{h}} P(\mathbf{Y}, \mathbf{E}=\mathbf{e}, \mathbf{H}=\mathbf{h})$$

hidden
variables

Their number is the number
of summations

Obvious problems:

- 1 Worst-case time complexity
- 2 Space complexity
- 3



Probability queries

Probability query

A **probability query** $\mathbf{P}(\mathbf{Y}|\mathbf{e})$ defines the posterior joint distribution of a set of **query variables** $\mathbf{Y}$ given specific values $\mathbf{e}$ for some **evidence variables**.

In principle, one could answer the query by **summing out**.

$$\mathbf{P}(\mathbf{Y}|\mathbf{E}=\mathbf{e}) = \alpha \mathbf{P}(\mathbf{Y}, \mathbf{E}=\mathbf{e}) = \alpha \sum_{\mathbf{h}} \mathbf{P}(\mathbf{Y}, \mathbf{E}=\mathbf{e}, \mathbf{H}=\mathbf{h})$$

Obvious problems: **n is the number of variables**

- 1 Worst-case time complexity $O(d^n)$ where d is the largest arity
- 2 Space complexity $O(d^n)$ to store the joint distribution
- 3 How to find the numbers for $O(d^n)$ entries???

Questions?